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Human robot collaboration in warehousing operations: a sociotechnical analysis

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Abstract: Warehouses are responsible for managing inventory levels, handling order fulfilment, and preparing items for transportation, and their efficiency determines logistics performance. Over the last few decades, warehouses have been integrating robotic solutions to enhance operational efficiency and reduce costs. However, the impact of these technologies on human workers and overall system performance remains underexplored, especially considering that human-robot systems are sociotechnical systems, where humans and robots interact with each other and work together to achieve a joint outcome. This study therefore systematically reviews existing literature on human-robot collaboration (HRC) in a warehouse context using a sociotechnical lens, employing the framework developed by Neumann et al. (2021), to map interactions between technology, tasks, human factors, and outcomes. Our findings indicate that robots can alleviate physical workload, but that they also introduce challenges related to mental and psychosocial factors. We identify research gaps related to humanoid robots and economic evaluations of robot introductions. By adopting a sociotechnical lens, this work provides insights for researchers by summarizing the state-of-the-art of HRC in warehouses and pointing out research gaps. In addition, it provides important insights obtained in research that can be considered in managerial decision-making, helping practitioners in optimizing HRC in warehouses by balancing human and machine strengths.

Keywords: Human robot interaction, Warehousing, Humanoid robot, Human factors, Industry 5.0

1. INTRODUCTION

To maintain a competitive position and meet highly customized demand, manufacturers and retailers are continuously seeking ways to reduce cost and improve process efficiency. In this context, logistics is crucial as it directly impacts production efficiency, customer satisfaction, and overall profitability (Tracey, 1998). Warehouses are critical elements of logistics systems, providing the physical space for organizing materials and preparing them for each step of the logistics chain. Efficient logistics relies on warehouses to manage inventory levels, handle order fulfilment, and prepare items for transportation (de Koster, 2023). Over the past few decades, companies have increasingly automated their warehouse operations by introducing various robotic solutions; however, end-to-end automation remains rare (Glock et al., 2021). Consequently, most modern warehouses adopt partial automation, where human workers perform tasks that are difficult to automate, such as item picking and shelf replenishment (Zhang et al., 2023). While partial automation alleviates some physically demanding tasks for workers, it often necessitates that the remaining tasks be performed manually and repetitively, particularly in areas like final inspection and packaging (Diego-Mas & Alcaide-Marzal, 2014). In this situation, human workers may have to repeat a limited set of tasks, increasing their movement intensity and workload (see a case example in production by Neumann et al. (2002)). Industry 4.0 (I4.0) systems are sociotechnical systems, which include social (human) and technical (machine) elements. This article considers collaborative robot (cobotic) systems using the sociotechnical perspective. The existing literature has focused on improving technical elements and lacks a detailed investigation of the impact of robot implementations on human workers and system outcomes from a sociotechnical lens (Panagou et al., 2024). Against this background, the following research question (RQ) is proposed: How does the adoption of robotic solutions affect human workers in a warehouse context from a sociotechnical perspective? To answer this RQ, we refer to the framework developed by Neumann et al. (2021) and conduct a systematic literature review, analysing how the existing literature designs and implements robotic solutions in warehouses from technology type, task scenario, human perceptions, and system outcomes perspectives. The remainder of the paper is structured as followings. Section 2 gives a background introduction about robots in warehouses, human factors, and the framework used for mapping the literature. Section 3 provides a summary of related literature reviews and positions our paper. Also, the literature selection process is presented. Section 4 presents the literature review results. Section 5 discusses the results and concludes the paper.

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2. CONCEPTUAL BACKGROUND

2.1 Robots in warehouse

Robots can automate material handling in warehouses and enhance operational efficiency. Autonomous mobile robots (AMRs), autonomous trucks, picking robots, and drones are commonly used robots in warehouses (de Koster, 2023). AMRs typically integrate light detection and ranging (LiDAR) and sensor fusion technology to map and locate themselves, and they are able to transport items along optimized routes (Sariff & Buniyamin, 2006). The most well-known AMRs are Robotic Mobile Fulfilment Systems, which use movable storage shelves that can be easily lifted by autonomous robots. These robots are self-organized and self-adaptive, and they can automatically find the best shelf based on its volume and transport it to the human worker for order picking (Wurman et al., 2008). E-commerce companies use such systems to decrease human labour costs. Autonomous trucks are another type of robot that can follow or be driven by the picker during order picking. Different from traditional industrial trucks, this type of truck can display pick information to guide the picker and follow the picker along the aisle autonomously (Koreis et al., 2023). Drones are also used in warehouses for inventory management, inspection and surveillance. They rely on indoor positioning systems or simultaneous localization and mapping to build a map of their surroundings and determine their location, and use a combination of sensors, cameras, and barcode readers or RFID scanners to identify and locate items within the warehouse (Maghazei et al., 2022).

2.2 Human factors

The terms human factors and ergonomics are often used interchangeably. The International Ergonomics and Human Factors Association defines human factors (HF) as the discipline of understanding interaction among humans and other elements of a system, applying theory, principles, data, and methods to design and optimize human wellbeing and overall system performance (IEA, 2024). HF involves physical, mental, perceptual, and psychosocial factors (Neumann & Dul, 2010). Physical factors focus on the biomechanical and physiological demands placed on the human body during task performance (Karwowski, 2005). Mental factors involve cognitive demands placed on individuals, including attention, memory, decision-making, problem solving, and information processing (Wogalter et al., 1998). Perceptual factors refer to the sensory demands involved in interacting with systems, including how individuals perceive, interpret, and respond to visual, auditory, tactile, or other sensory information (Wogalter et al., 1998). Psychosocial factors encompass interpersonal relationships and support, workplace culture, job demands and stress levels that influence individuals' behaviour, emotions, and health (Carayon, 2006). Human workers excel at decision-making and handling unpredictable situations. They are essential in partially automated warehouses due to their motor and cognitive capabilities that robots cannot replace yet (Sgarbossa et al., 2020). As human workers are integral to operations, their performance directly affects overall system efficiency and outcomes. Despite its significance, HF has often been overlooked in human-robot system design (Panagou et al., 2024) and warehouse work more generally (Grosse et al., 2017).

2.3 Mapping framework

Human-robot systems can be understood as an ecosystem consisting of interdependent technology, a personnel subsystem, and organization. The robot and personnel components are complementary, and together they determine the social and psychological workplace demands and performance (Carayon, 2006). Neumann et al. (2021) proposed a systematic framework to consider HF in the implementation of new technologies in operations systems in an I4.0 context. Their framework is applied in five steps, which are 1) define the technology, 2) determine the affected humans, 3) identify task scenarios, 4) analyse task and its impacts, 5) analyse outcome. The developed framework allows a systematic assessment of the impacts of technology adoption on humans, task, and system performance from a sociotechnical perspective. In this study, we utilize this framework in a literature review of warehouse research and provide a comprehensive mapping of how the introduction of robots in warehouses impacts humans, tasks, and system performance (Figure 1).

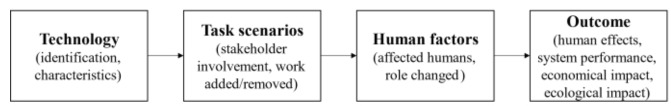

Figure 1: Mapping framework of robot implementation and their impacts in a warehouse context

3. METHODOLOGY

3.1 Existing literature reviews

To differentiate our study from existing literature reviews, we give an overview of related literature reviews that discuss the implementation of robots in a production or logistics context. For example, Liu et al. (2024) and Keshvarparast et al. (2023) investigated the effect of cobots on workers' safety, human-robot task scheduling, and workplace design in production, with their reviews paying little attention to HF. De Simone et al. (2022) identified the key factors impacting the performance of operators in human-robot collaboration (HRC), with a focus on safety, physical ergonomics and productivity. Simões et al. (2022) classified the literature into different categories, providing guidelines to facilitate HRC by considering both technology characteristics and human demands. Storm et al. (2022) analysed the factors affecting workers' physical and mental health in HRC and concluded that most studies still focus on risk assessment and physical safety. Most related to us are the studies of Panagou et al. (2024) and Lorson et al. (2023). Panagou et al. (2024) examined how the design features of robots affect human operators and found that design features are highly interactive with the outcome of human perception. Lorson et al. (2023) is the only study concentrating on logistics that identified and analysed relevant behaviour issues for HRC across various warehousing activities. However, they conducted a narrative review and did not systematically map the technology-human-task-outcome relationship.

While existing studies provide valuable insights, they focus either on technological aspects or human factors in isolation. A comprehensive sociotechnical perspective that systematically examines the interdependencies between technology, tasks, HF, and outcomes in HRC remains underexplored, especially in a warehousing context. The study at hand differs from the existing literature by identifying the technology characteristics and their impact on task scenarios, HF, and system performance with a focus on warehousing.

3.2 Literature search process

We adopted a four-step approach to develop the literature sample: (I) identify groups of keywords, (II) search scientific databases, (III) screen the literature, and (IV) complement the literature sample using a snowball search (Cooper, 2017). First, we identified three groups of keywords based on the mapping framework (section 2.3) and related reviews (section 3.1). Table 1 lists the identified keywords. Cluster A keywords are related to human-robot-interaction, and Cluster B keywords are related to warehouses. Cluster C keywords are related to HF. In the next step, we used the Boolean operator 'AND' to combine the keywords from the three clusters to generate the search string and conducted a search in the Web of Science database, which is a scholarly leading database (Birkle et al., 2020). Studies published until the October 15th, 2024, and featuring at least one of the keywords in the article title, abstract, or list of keywords were retrieved for further analysis. The initial literature sample consisted of 72 papers. In the next step, we screened the title, abstract, and source of the articles and selected those papers that 1) examine HRC and consider HF in their study, 2) focus on a logistics warehouse context. In this stage, we excluded 41 papers. Next, the screened papers were downloaded for full reading. 13 papers were included in the literature sample in this step. Finally, a snowball search was conducted to identify additional papers, with two additional papers resulting from this search. The final literature sample consists of 15 papers.

Table 1. Keywords used for literature search

Cluster	Keywords
A	"human robot collaboration", "human robot interaction", "human robot interface", "human machine collaboration", "human machine interaction", "human machine interface", "collaborative robot*", "cobot*", "service robot*", "automated guided vehicle*", "AGV", "goods-to-person robot*", "g2p robot*", "kiva robot*", "humanoid*"
B	"logistics", "warehous*", "order pick*, "material* handling", "inventory management", "distribution center*", "packaging", "transportation", "delivery"
C	"human factor*", ergonomics

4. RESULTS

This section reviews the literature sample according to the type of robots adopted and the main tasks they are carrying out in warehouses. We focus on how the adoption of different types of robots impacts task scenarios and HF as well as the outcome.

4.1 Autonomous materials transportation

In the reviewed papers, AGVs are frequently used for intralogistics materials supply, aiming to completely replace human workers for transporting pallets (Afonso et al., 2021; Chen et al., 2020; Kopp et al., 2023; Sirintuna et al., 2024; Vermin et al., 2024).

In most circumstances, the role of human workers does not change, i.e., AGVs are introduced to support human workers in completing their own task, for example order picking and materials handling. One exception is the study of Afonso et al. (2021), where AGVs replaced forklift drivers for conducting milk runs and manoeuvring containers. In this case, there is no physical workload for the forklift driver, as no human is needed for handling containers. Vermin et al. (2024) pointed out that introducing robots may induce workload imbalance at picking stations, as some picking stations may have heavy and high volumes of items, while other stations may have less physically demanding picking tasks, depending on the item distribution policy. Besides addressing physical workload, Chen et al. (2020) showed that order picker's mental workload increased when they interact with AGVs, as measured by pupil diameter increases. Sirintuna et al. (2024) found that human workers perceive less mental workload when they receive a combination of feedback (virtual fixture and vibrotactile) during their interaction with robots compared to receiving no feedback. Kopp et al. (2023) found that operational staff (production and logistics workers) and managerial staff (managers and project leaders) perceive robot adoption differently. Logistics workers are less satisfied with the AGV implementation and more concerned about losing their jobs than production workers, while managerial staff is more satisfied with the performance of AGVs than operational staff.
Regarding outcomes, Chen et al. (2020) showed that there is a slight human productivity deficit due to additional attention spent on scanning the surroundings to localize robots, while the decrease in human productivity is compensated by the relatively large gain from the robot's output, leading to an increase in system performance. Afonso et al. (2021) showed that introducing AGVs can reduce the materials supply time and increase operational performance due to its on-going operation in the entire 8-hour shift. Besides addressing system performance, their results showed that introducing AGVs can reduce costs associated with direct labour, accounting for about 14% of the total operational cost

4.2 Robot-assisted manual order picking

Autonomous trucks move along with the order picker, aiming to guide order pickers by demonstrating order list when they adopt a leading role, or they simply transport order crates when working as a servant. In this HRC scenario, the role of the order picker does not change. They are still responsible for picking items; however, a part of their work is removed, such as steering the truck, pushing the cart, or searching item locations, as the autonomous truck can lead the order picking process and reduce the physical workload of order pickers (Koreis et al., 2023; Pasparakis et al., 2023). To successfully introduce autonomous trucks, psychosocial factors need to be considered. Pasparakis et al. (2023) argued that the overall job satisfaction of order pickers is linked to perceived ease of use of the robot, as well as to the added value they believe the human-robot-collaboration introduces. Therefore, engaging order pickers in the beginning of robot introduction is crucial to let them gain trust in the robot. Also, team leaders need to be engaged in providing training to order pickers in terms of how the collaborative tasks can be completed efficiently (Lambrechts et al., 2021). In addition, social group dynamics need be considered because people might envy those among a group of peers who receive new technology equipment like autonomous truck, whereas others still have to work with older technology (Koreis et al., 2023).

Compared to traditional trucks, using autonomous trucks is estimated to reduce the time required to complete the order picking task by 33.6% (Koreis et al., 2023). However, such robots are expensive (€40-90k), and the required investment volume can only be afforded by a few companies. Thus, evaluating whether the initial investment can deliver the desired output and compensate the savings in staff must be evaluated from an economic perspective (Lambrechts et al., 2021).

4.3 Partial or fully automated order picking

Autonomous picking robots are deployed in warehouses to complete picking task automatically. One variant is hybrid order picking, where autonomous picking robots work together with human order pickers, with the robots being responsible for picking certain types of items (Proia et al., 2022; Zhang et al., 2023). The other variant is the fully automated picking system, where the order picking task is fully delegated to picking robots. In a hybrid order picking scenario, the role of order pickers does not change, but is restricted to the picking of certain items (Zhang et al., 2023) or simply performing palletizing tasks (Proia et al., 2022). In the fully automated picking scenario, a new role is required that is responsible for training the automated picking system, such that the object detection algorithms used by the robots can be adapted to new tasks in a dynamically changing environment (Rieder et al., 2021).

Zhang et al. (2023) considered human energy expenditure in a simulated hybrid picking scenario. Their results show that implementing autonomous picking robots can reduce energy expenditure by allocating robots to pick items with longer traveling distances and assigning human pickers to travel less and pick more items. They concluded that hybrid order picking is preferred if a high daily throughput is required. Proia et al. (2022) used drones to pick items and deliver them to humans for palletizing. They assessed order pickers' physical workload when collaborating with drones and tried to optimize the item handling process, such that human workload is minimized upon receipt of items from drones. In fully automated picking, although order pickers do not need to perform manual order picking anymore, humans are needed to train the picking system. Rieder et al. (2021) estimated that this translates into around 158 hours in training for the first year. The time invested in training is compensated by lower utilization of labour, resulting in approximately 256 hours saved for each hour invested in training.

All studies conclude that introducing autonomous picking robots improves system performance compared to manual order picking. From an economic perspective, Rieder et al. (2021) calculated net present value and return on investment and concluded that introducing autonomous picking robot is worth the investment.

4.4 Cooperation with cobots at the workbench

Cobots can be adapted to stable or mobile workbenches to assist workers in completing various tasks, such as co-transportation (Fruggiero et al., 2020) and packaging (Accorsi et al., 2019). Accorsi et al. (2019) showed that using cobots to perform primary and secondary packaging, while letting human workers conduct inspection and setup, will reduce ergonomic load of the workers. Workers may have different perceptions when they interact with cobots. For example, Rossato et al. (2021) showed that older workers (>55) perceived cobots as slightly less useful than younger workers (35-54); however, both groups have a high acceptance level for the cobot.

In terms of outcome, Accorsi et al. (2019) concluded that investing in a packaging cobot can pay off in around 2-2.5 years, depending on the production lot size and the number of lots processed per day.

5. DISCUSSION AND CONCLUSION

This study systematically examined the adoption of robots in warehouse operations through a sociotechnical lens, focusing on the interplay between technology, task scenarios, human factors, and system outcomes. First, we found that robots with different automation levels are adopted in warehouses, for example autonomous robots that assist humans in manual order picking, cobots that replace human workers to perform heavy and repetitive tasks, and partial and fully automated robots that conduct order picking. However, we did not find any literature that investigates the implementation of humanoid robots, even though we included humanoid-related keywords in our search string. Given that humanoids can grab objects with their flexible fingers and perform tasks at a relatively low cost (Malik et al., 2023), there is potential to use this type of robot for fine-motor operations, such as goods picking and inspection. We thus encourage future studies to explore the potential of humanoids in warehousing activities. In terms of HF, our findings show that in most circumstances, human workers' roles do not change, i.e., they still perform their routine work, but parts of their tasks are delegated to robots. Indeed, most studies addressed the reduction of physical workload. However, how changes in tasks impact human workers and whether this leads to workload intensification needs to be better understood. For example, assigning robots to complete transportation faster may oblige humans to lift more and pick faster in the meantime. In addition, human workers face learning-, adaptation- and trust-related issues in the interaction, and they are only satisfied when they feel safe with the interaction and perceive that the robots can add value to their collaborative tasks (Pasparakis et al., 2023). In this regard, we encourage future works to evaluate the impact of robots on mental, perceptual, and psychosocial factors. In addition, the existing literature does not clearly address what types of works should be conducted by humans when robots take over a part of their work. We thus encourage future studies to evaluate the design of new forms of warehouse work, such that human capabilities (e.g., cognitive skills, decision-making, creativity) and machine strengths (precision, repeatability, productivity) can be optimally merged (Hoc, 2000). Our review showed that the introduction of robots can increase system performance, while only a few studies mentioned economic impacts. We encourage future studies to further evaluate the return on investment for the introduction of robots, considering factors such as lifespan, reliability, maintenance costs, and cost of a backup system (e.g., human resources).

While this study provides valuable insights into the adoption of robots in warehouses, several limitations should be acknowledged. First, we only used Web of Science for the literature search, with other databases such as Scopus not being included, which may have resulted in some literature being missed. Second, we might have missed some keywords during the database search. However, we conducted a snowball search to complement the literature search and to address this limitation.

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